

Project : Derivable Judgement

Part A – Theory :-



1. What is Inferential Statistics?

- Inferential Statistics is used to make decisions about a whole population using sample data.
It helps us predict, estimate, and test hypotheses.



2. What is Hypothesis Testing and its Components?

- Hypothesis Testing is a statistical method used to check whether a claim is true.

Components:

- H_0 (Null Hypothesis): No effect or no difference.
- H_1 (Alternative Hypothesis): Effect or difference exists.
- Significance Level (α): Usually 0.05.
- Test Statistic: Value calculated from sample data.
- Decision Rule: Accept or Reject H_0 .

3. Confidence Interval and Critical Value

- **Confidence Interval (CI):** Range in which the true population value is likely to exist.
- **Critical Value:** Boundary value used to decide whether H_0 should be rejected.

4. What is a p-value?

- A p-value measures how strongly the sample data supports H_0 .
 - ◆ $p < 0.05 \rightarrow$ Reject H_0
 - ◆ $p \geq 0.05 \rightarrow$ Accept H_0

5. Type I and Type II Errors

- **Type I Error (α):**
Rejecting a true H_0 .
- **Type II Error (β):**
Accepting a false H_0 .

6. z-test, t-test, Chi-square test, ANOVA

- **z-test:** Compare means when sample size is large.
- **t-test:** Compare means when sample size is small.
- **Chi-square Test:** Checks relationship between categorical variables.

→ **ANOVA:** Compares means of more than two groups.



7. What is Covariance?

→ Covariance shows how two variables move together.

- Positive → Increase together
- Negative → One increases while the other decreases



8. What is Correlation?

→ Correlation measures strength and direction of relationship between variables.

→ Range:

- +1 → Perfect Positive
- 0 → No Relation
- -1 → Perfect Negative